

Morphological and genomic characterization of a novel *Hepatozoon* lineage in the critically endangered insular pitviper *Bothrops sazimai* (Squamata: Serpentes: Viperidae)

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Introduction

Hepatozoon (Apicomplexa: Adeleorina: Hepatozoidae) is the blood parasite most often reported in snakes. Most species were described from gamont shape and short 18S rRNA fragments. Recent studies in South America keep finding lineages that morphology alone cannot separate.

Bothrops sazimai lives only on Ilha dos Franceses, Espírito Santo, Brazil (Fig. 1). Rising sea levels cut the island off from the mainland within the last 11,000 to 55,000 years². Its describers proposed it as Critically Endangered¹. Instituto Butantan keeps a captive colony for conservation, and health screening of these snakes also reveals their parasites.

Objective. Characterize a *Hepatozoon* lineage found in *Bothrops sazimai* by gamont morphology, 18S rRNA, and complete mitochondrial and apicoplast genomes, and compare it with known species.



Figure 1. *Bothrops sazimai* from Ilha dos Franceses. Photo: Ricardo Sawaya.

Methods

We drew blood from an adult female *Bothrops sazimai* of the Instituto Butantan colony and compared its gamonts with 28 records of snake *Hepatozoon* from 14 studies. We pulled parasite reads out of the host's Oxford Nanopore genome assembly (Fig. 2) and placed the 18S variants in a maximum likelihood tree of 150 adeleorin sequences.

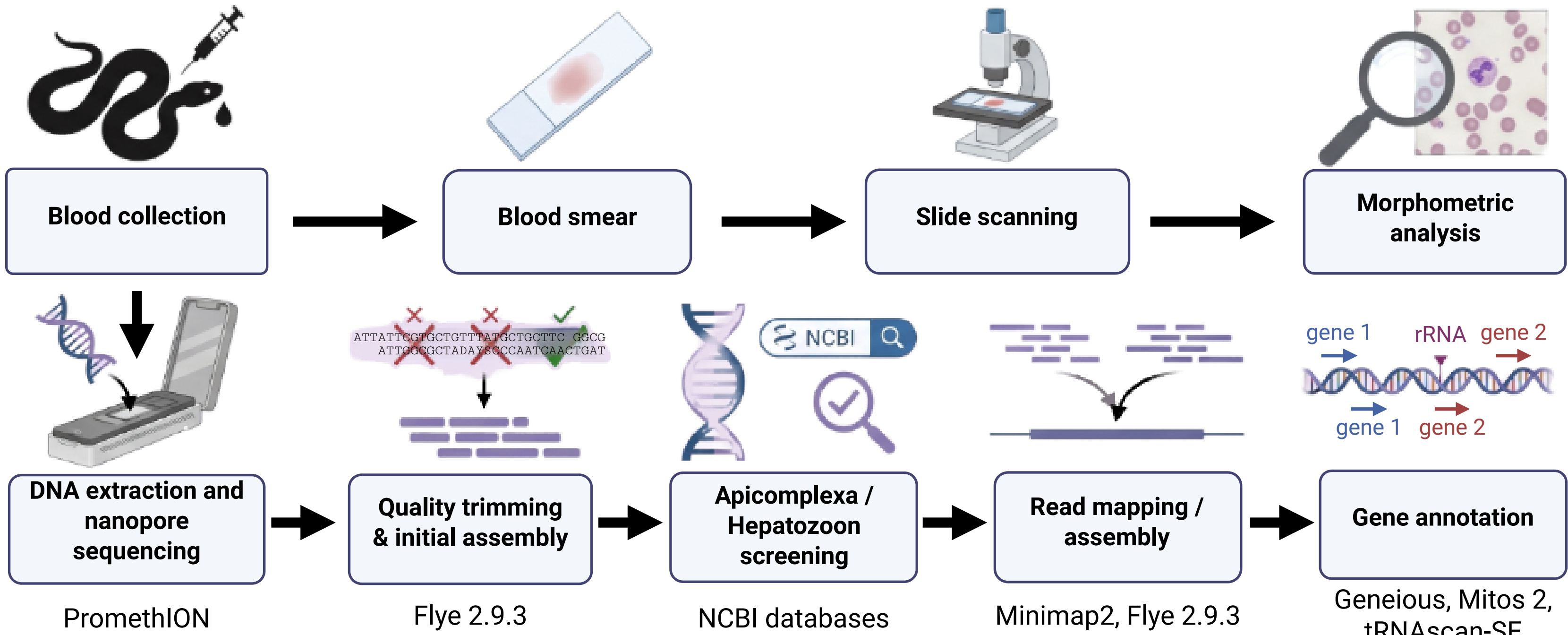


Figure 2. Workflow: gamont morphometry (top) and parasite genomics (bottom). Created in BioRender (BioRender.com).

Results

Short gamonts with a long, narrow nucleus

Gamonts measured $11.9 \pm 1.3 \mu\text{m}$ by $4.3 \pm 0.7 \mu\text{m}$ (Fig. 3).

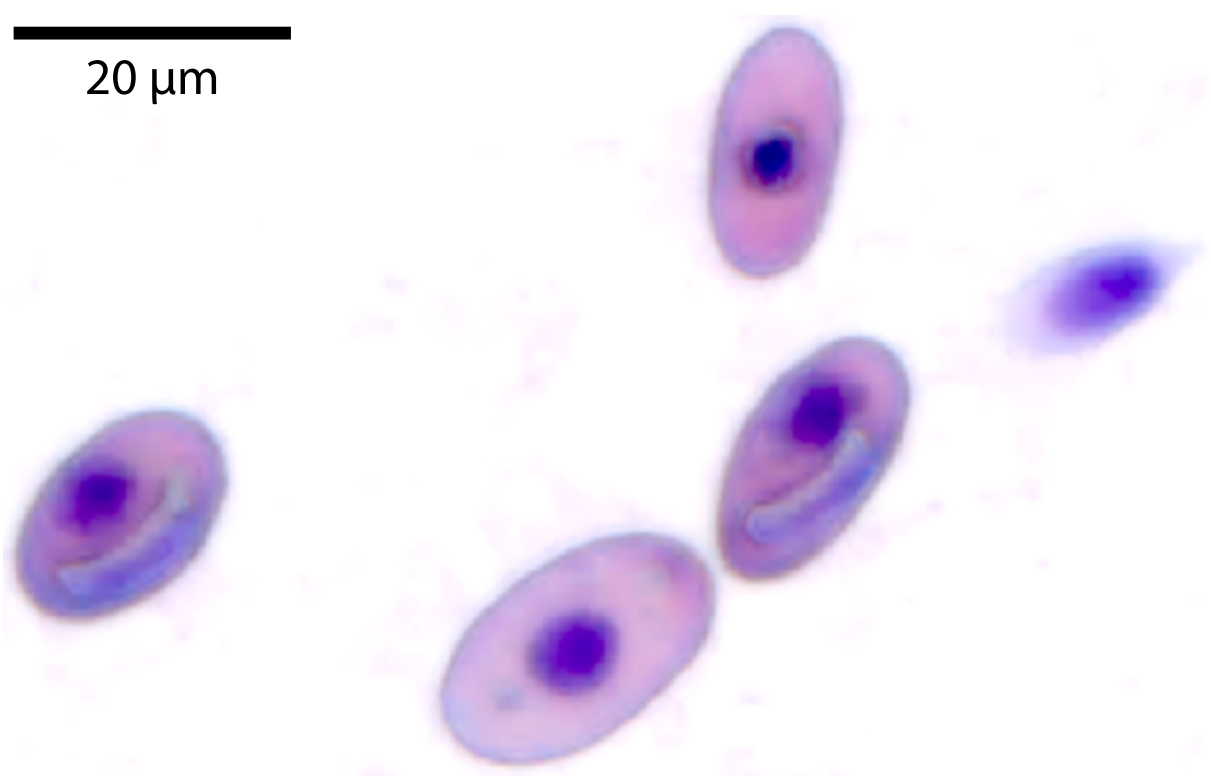


Figure 3. Blood smear of *Bothrops sazimai*. Two erythrocytes carry curved *Hepatozoon* gamonts beside the host nucleus.

Only *H. atrocis* has shorter gamonts among the compared species (Fig. 4A). The gamont is short and broad (length/width 2.8), but its nucleus is long and narrow (length/width 2.1). No compared species combines the two (Fig. 4B).

Two 18S variants that do not group together

The host carries two 18S rRNA variants (944 and 271 reads). Both fall inside *Hepatozoon* (Fig. 5). V2 sits in a clade of Neotropical snake parasites (SH-aLRT 97.7, UFBoot 100), next to *H. annulatum*. V1 falls outside that clade, as sister to a group of snake and turtle parasites.

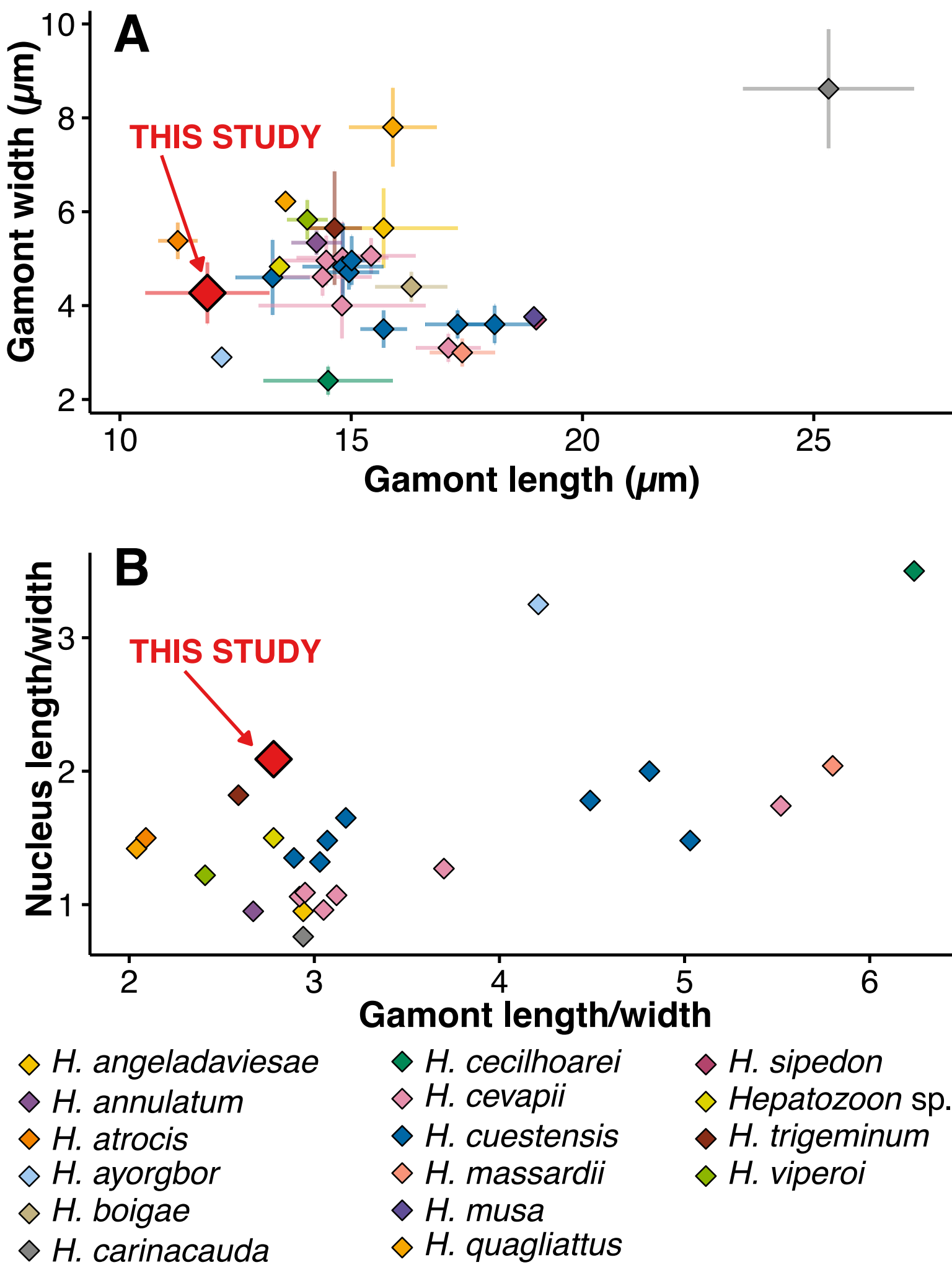


Figure 4. Gamonts of snake *Hepatozoon*. A: size (mean $\pm$ SD). B: gamont and nucleus shape.

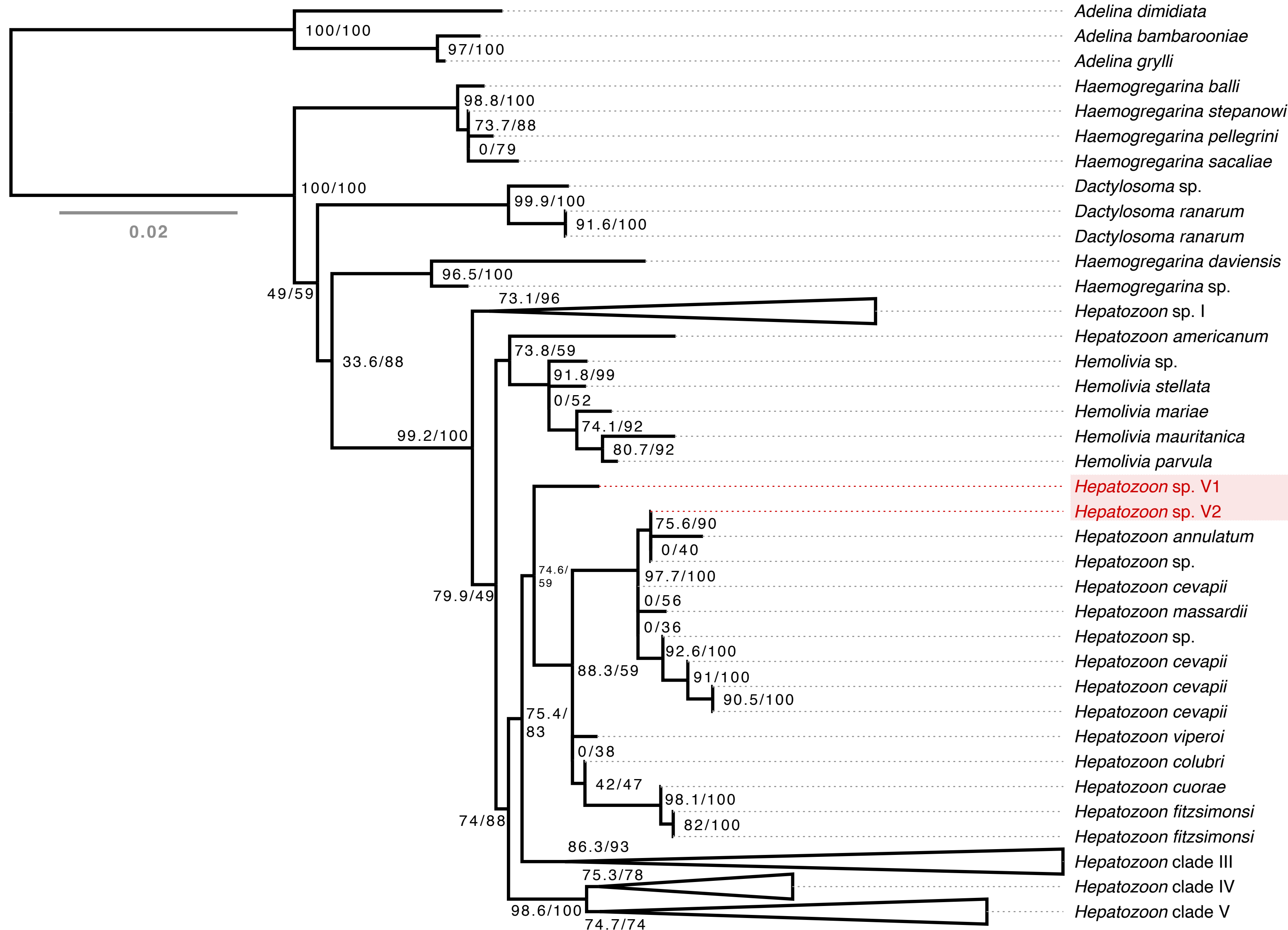


Figure 5. Maximum likelihood tree of 150 adeleorin 18S rRNA sequences, rooted on *Adelina*. Node labels: SH-aLRT/ultrafast bootstrap. Triangles: collapsed clades. Red: V1 and V2, the two variants from *Bothrops sazimai*. Scale bar: substitutions per site.

A large mitochondrial genome with a new gene order

The mitochondrial genome is 10,479 bp, 63% longer than the largest published congener (6,413 bp). It carries *cox1*, *cox3*, *cob*, and 25 rRNA fragments, like its relatives. The extra length sits between *cox1* and *cob*, where *cox3* now follows *cob* (Fig. 7). The apicoplast genome is 32,316 bp, with 22 protein-coding genes, 26 tRNA genes, and 4 rRNA genes (Fig. 6).

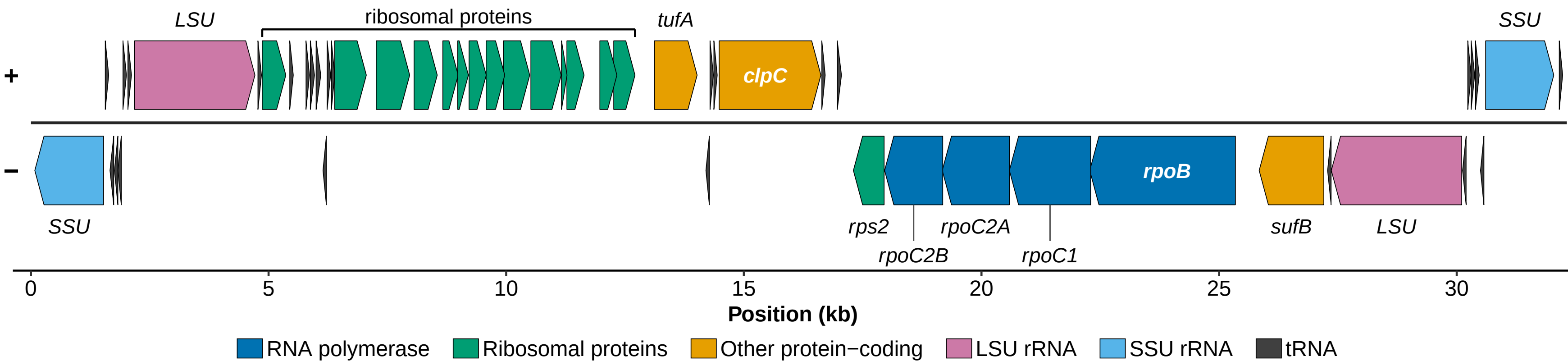


Figure 6. Apicoplast genome of *Hepatozoon* sp. from *Bothrops sazimai*, circular and drawn linear. Genes on the + strand sit above the line, genes on the - strand below.

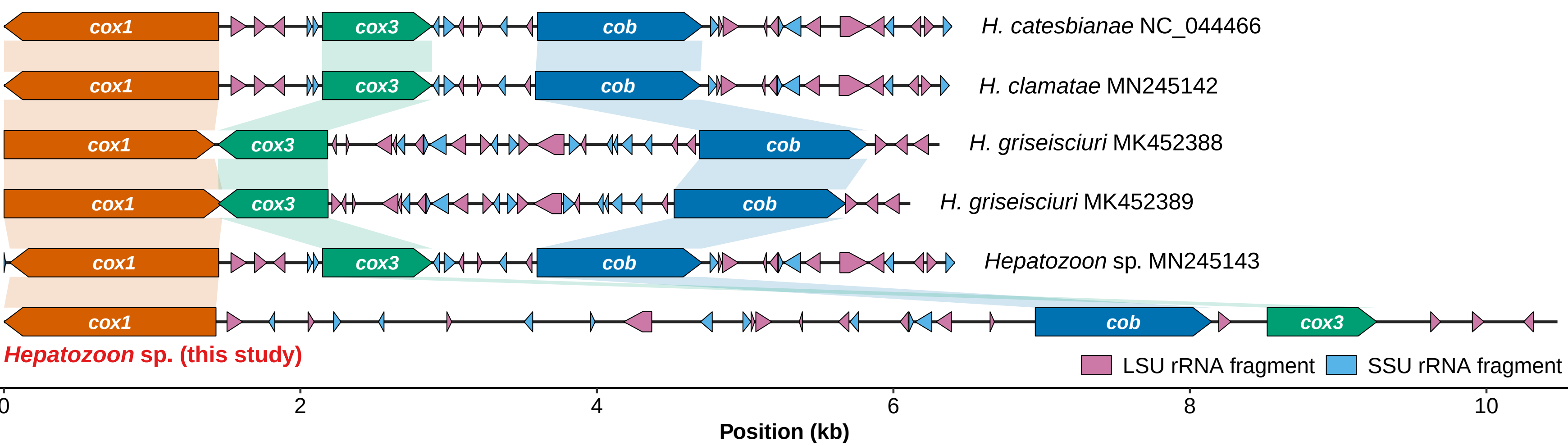


Figure 7. Mitochondrial genomes of *Hepatozoon*. Ribbons link *cox1*, *cox3*, and *cob* between neighboring genomes.

Conclusions

- *Bothrops sazimai* carries a *Hepatozoon* lineage that differs from every compared species in gamont shape, 18S rRNA, and mitochondrial genome.
- Its mitochondrial genome is the largest reported for the genus, with a gene order not seen before.
- Two 18S variants in one host point to divergent rDNA copies or a mixed infection.
- Host genome projects also sequence blood parasites, at no extra cost.

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References. ¹Barbo FE et al. 2016. Zootaxa 4097(4): 511-529. ²Barbo FE et al. 2022. Systematics and Biodiversity 20(1): 1-25.

